RetroDiff: Training Objective Derivation

Let x denote a clean data sample and x_t its noised version at diffusion step $t \in \{1, \ldots, T\}$. The forward process $q(x_{1:T}|x)$ gradually corrupts x; the model p_θ learns to reverse it. Applying Jensen's inequality to $\log p_\theta(x)$ yields the variational lower bound (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)}\right] = -\mathcal{L}_{\text{ELBO}} \tag{1}$$

The bound in (1) is tight when the variational posterior matches the true reverse process. Expanding the joint $p_\theta(x_{0:T})$ via the Markov factorisation and grouping terms by diffusion step decomposes $\mathcal{L}_{\text{ELBO}}$ into a reconstruction term and a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0|x_1)]}_{\text{reconstruction}} + \sum_{t=2}^{T} \underbrace{\mathrm{KL}(q(x_{t-1}|x_t, x) \,\|\, p_\theta(x_{t-1}|x_t))}_{\text{step-}t\text{ denoising}} \tag{2}$$

Each KL term in (2) measures how well the learned reverse step $p_\theta(x_{t-1}|x_t)$ approximates the tractable posterior $q(x_{t-1}|x_t, x)$. **RetroDiff**'s key contribution is to condition every denoising step on a set $R(x) = \{r_1, \ldots, r_k\}$ of k exemplars retrieved from the training corpus based on x. The per-sample ELBO is evaluated under this conditioning, and the final training objective averages over the data distribution:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}}\big[\mathcal{L}_{\text{ELBO}}\big(x \,\big|\, R(x)\big)\big] \tag{3}$$

Minimising (3) encourages the model to leverage retrieved exemplars as soft structural priors at every denoising step, without modifying the diffusion schedule or the ELBO derivation itself.